

MINISTRY OF EDUCATION

Databases

Laboratory Work 1

ER Diagram of the International Airport Database System

Submission format: PDF document containing ER-Diagram
of the system and textual description of the solution.

System: **International Airport Database**

Schema: **airport** (PostgreSQL)

Entities: 10 | Relationships: 11 | Primary Keys: 10

1. Identify Entities

Based on the system description of the International Airport Database, the following **10 entities** were identified:

1. **Airline** — Airline companies that operate flights.
2. **Airport** — Airports involved in departures and arrivals.
3. **Flight** — Scheduled and actual flight information.
4. **Passenger** — Persons who book and travel on flights.
5. **Booking** — Reservation of a seat on a specific flight by a passenger.
6. **Boarding Pass** — Document issued for a confirmed booking (seat assignment).
7. **Baggage** — Luggage registered against a booking.
8. **Baggage Check** — Security/inspection result for baggage of a booking.
9. **Security Check** — Airport security screening result for a passenger.
10. **Booking Change** — Audit record of modifications made to a booking.

2. Determine Attributes

For each entity the attributes are listed below. Primary keys (PK), foreign keys (FK), data types, and constraints (NOT NULL, UNIQUE, DEFAULT, CHECK) are indicated.

2.1 Airline

AIRLINE		
Attribute	Data Type	Constraints / Notes
airline_id (PK)	SERIAL	PRIMARY KEY, NOT NULL
airline_code	VARCHAR(10)	NOT NULL, UNIQUE
name	VARCHAR(100)	NOT NULL
country	VARCHAR(50)	NOT NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

2.2 Airport

AIRPORT		
Attribute	Data Type	Constraints / Notes
airport_id (PK)	SERIAL	PRIMARY KEY, NOT NULL
airport_name	VARCHAR(100)	NOT NULL
country	VARCHAR(50)	NOT NULL
state	VARCHAR(50)	NULL allowed
city	VARCHAR(50)	NOT NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

2.3 Flight

FLIGHT		
Attribute	Data Type	Constraints / Notes
flight_id (PK)	SERIAL	PRIMARY KEY, NOT NULL
departing_gate	VARCHAR(10)	NULL allowed
arriving_gate	VARCHAR(10)	NULL allowed
airline_id (FK)	INT	NOT NULL → airline(airline_id)
departure_airport_id (FK)	INT	NOT NULL → airport(airport_id)
arrival_airport_id (FK)	INT	NOT NULL → airport(airport_id)

scheduled_departure_time	TIMESTAMP	NOT NULL
scheduled_arrival_time	TIMESTAMP	NOT NULL
actual_departure_time	TIMESTAMP	NULL allowed
actual_arrival_time	TIMESTAMP	NULL allowed
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

2.4 Passenger

PASSENGER		
Attribute	Data Type	Constraints / Notes
passenger_id (PK)	SERIAL	PRIMARY KEY, NOT NULL
first_name	VARCHAR(50)	NOT NULL
last_name	VARCHAR(50)	NOT NULL
gender	VARCHAR(10)	NULL allowed
date_of_birth	DATE	NULL allowed
country_of_citizenship	VARCHAR(50)	NULL allowed
country_of_residence	VARCHAR(50)	NULL allowed
passport_number	VARCHAR(20)	UNIQUE (nullable)
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

2.5 Booking

BOOKING		
Attribute	Data Type	Constraints / Notes
booking_id (PK)	SERIAL	PRIMARY KEY, NOT NULL
flight_id (FK)	INT	NOT NULL → flight(flight_id)
passenger_id (FK)	INT	NOT NULL → passenger(passenger_id)
status	VARCHAR(20)	NOT NULL, DEFAULT 'confirmed'
booking_platform	VARCHAR(50)	NULL allowed
ticket_price	DECIMAL(10,2)	NOT NULL
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

2.6 Boarding Pass

BOARDING_PASS		
Attribute	Data Type	Constraints / Notes
boarding_pass_id (PK)	SERIAL	PRIMARY KEY, NOT NULL
booking_id (FK)	INT	NOT NULL, UNIQUE → booking(booking_id) [1:1]
seat	VARCHAR(10)	NOT NULL
boarding_time	TIMESTAMP	NULL allowed
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

2.7 Baggage

BAGGAGE		
Attribute	Data Type	Constraints / Notes
baggage_id (PK)	SERIAL	PRIMARY KEY, NOT NULL

booking_id (FK)	INT	NOT NULL → booking(booking_id)
weight_kg	DECIMAL(5,2)	NOT NULL, CHECK (weight_kg > 0)
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

2.8 Baggage Check

BAGGAGE_CHECK		
Attribute	Data Type	Constraints / Notes
baggage_checking_id (PK)	SERIAL	PRIMARY KEY, NOT NULL
booking_id (FK)	INT	NOT NULL → booking(booking_id)
passenger_id (FK)	INT	NOT NULL → passenger(passenger_id)
check_results	VARCHAR(100)	NULL allowed
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

2.9 Security Check

SECURITY_CHECK		
Attribute	Data Type	Constraints / Notes
security_check_id (PK)	SERIAL	PRIMARY KEY, NOT NULL
passenger_id (FK)	INT	NOT NULL → passenger(passenger_id)
check_results	VARCHAR(100)	NULL allowed
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

2.10 Booking Change

BOOKING_CHANGE		
Attribute	Data Type	Constraints / Notes
change_id (PK)	SERIAL	PRIMARY KEY, NOT NULL
booking_id (FK)	INT	NOT NULL → booking(booking_id)
change_description	TEXT	NULL allowed
changed_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP

3. Identify Relationships

Relationships were derived from the foreign-key constraints and the business rules described in the system specification. Cardinality and participation are stated for each.

#	Relationship	Cardinality	Participation / Notes
1	Airline operates Flight	1 : N	One airline operates many flights. Flight must belong to an airline (total participation).
2	Airport is departure of Flight	1 : N	One airport can be the departure airport of many flights. Flight must have a departure airport.
3	Airport is arrival of Flight	1 : N	One airport can be the arrival airport of many flights. Flight must have an arrival airport.
4	Flight has Booking	1 : N	One flight can have many bookings. Booking must reference a flight.
5	Passenger makes Booking	1 : N	One passenger can make many bookings. Booking must reference a passenger.
6	Booking issues Boarding Pass	1 : 1	Each booking has at most one boarding pass (UNIQUE on booking_id). Boarding pass requires a booking.
7	Booking has Baggage	1 : N	One booking can register many baggage items. Baggage must belong to a booking.
8	Booking has Baggage Check	1 : N	One booking can have multiple baggage checks. Check requires a booking.
9	Passenger has Baggage Check	1 : N	One passenger can appear in many baggage checks. Check requires a passenger.
10	Passenger passes Security Check	1 : N	One passenger can have many security checks. Check requires a passenger.
11	Booking has Booking Change	1 : N	One booking can have many change records (audit log). Change requires a booking.

Note on Many-to-Many: No pure many-to-many relationship is modelled with a separate linking table in this schema. The associative nature of Booking (Passenger–Flight) and Baggage Check (Booking–Passenger) already resolves potential M:N associations.

4–7. ER Diagram

The diagram below was produced according to the identified entities, attributes and relationships. Entity boxes contain primary keys (PK, bold blue) and foreign keys (FK, red). Relationship lines are labelled with cardinality (1 or N) at each end.

ER Diagram — International Airport Database (LAB1)

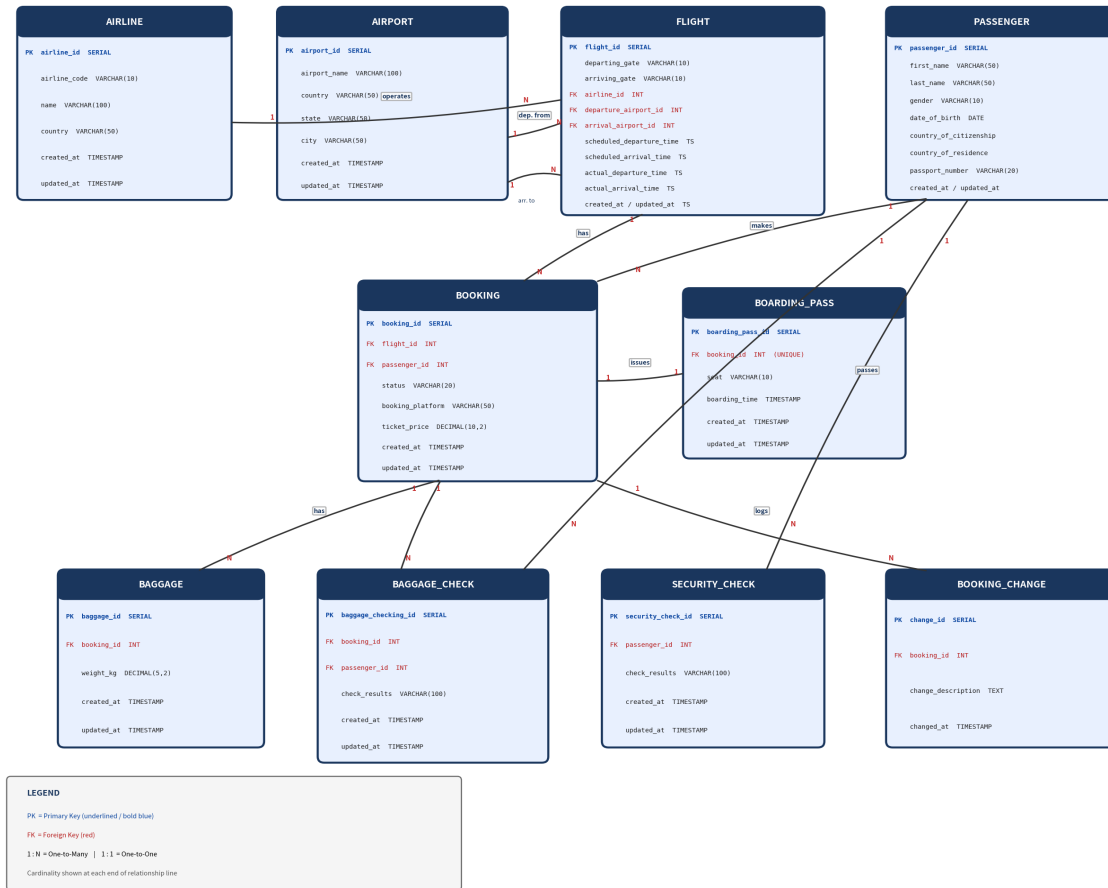


Figure 1. Entity-Relationship Diagram of the International Airport Database

8. Document the Diagram — Legend and Description

8.1 Notation Legend

Symbol / Notation	Meaning
Rectangle (entity box)	An entity (table) of the database.
Dark header bar	Entity name.
PK (blue, bold)	Primary key attribute. Uniquely identifies each row.
FK (red)	Foreign key attribute. References a primary key of another entity.
Line between boxes	Relationship between two entities.
1 : N	One-to-Many cardinality.
1 : 1	One-to-One cardinality (boarding_pass.booking_id is UNIQUE).
SERIAL	Auto-incrementing integer (PostgreSQL).
TIMESTAMP / DATE / DECIMAL / VARCHAR / TEXT	PostgreSQL data types used for attributes.

8.2 Brief Description of Each Entity

- Airline** — Represents an airline company. Identified by airline_id. airline_code is unique (e.g. SU, BA).
- Airport** — Physical airport location. Identified by airport_id. Used both as departure and arrival airport of flights.
- Flight** — A single scheduled (and optionally actual) flight operated by one airline between two airports.
- Passenger** — A person who books flights. passport_number is unique when provided.
- Booking** — A reservation linking one passenger to one flight. Holds status, platform and ticket price.
- Boarding Pass** — Issued for a booking (1:1). Contains seat number and boarding time.
- Baggage** — A piece of luggage belonging to a booking. Weight must be positive.
- Baggage Check** — Result of inspecting baggage for a given booking and passenger.
- Security Check** — Result of the airport security screening performed on a passenger.
- Booking Change** — Audit-log entry describing a change made to a booking (history of modifications).

8.3 Integrity and Design Decisions

- All foreign keys are declared NOT NULL, enforcing total participation of the child side in each relationship (except optional attributes such as actual times, gates, gender, etc.).
- Boarding Pass has a UNIQUE constraint on booking_id, implementing a true 1:1 relationship with Booking.
- Baggage.weight_kg has a CHECK constraint (weight_kg > 0) to prevent invalid data.
- created_at / updated_at columns with DEFAULT CURRENT_TIMESTAMP provide basic audit timestamps on almost every entity.
- Booking Change is a separate entity so that the history of modifications is preserved without overwriting the current booking record.
- The schema is placed in the PostgreSQL schema **airport** (CREATE SCHEMA airport).